COMP2111 Database Design

Class Practice 2

The following tables form part of a database held in a relational DBMS.

Hotel (hotelNo, hotelName, city)

Room (roomNo, hotelNo, type, price)

where hotelNo is a FK referring to Hotel(hotelNo)

To get details of hotelName

Guest (guestNo, guestName, guestAddress)

Booking (hotelNo, guestNo, dateFrom, dateTo, roomNo)

where hotelNo is a FK referring to Hotel(hotelNo)

guestNo is a FK referring to Guest(guestNo)

(roomNo, hotelNo) is a FK referring to Room(roomNo, hotelNo)

To get details of guestName

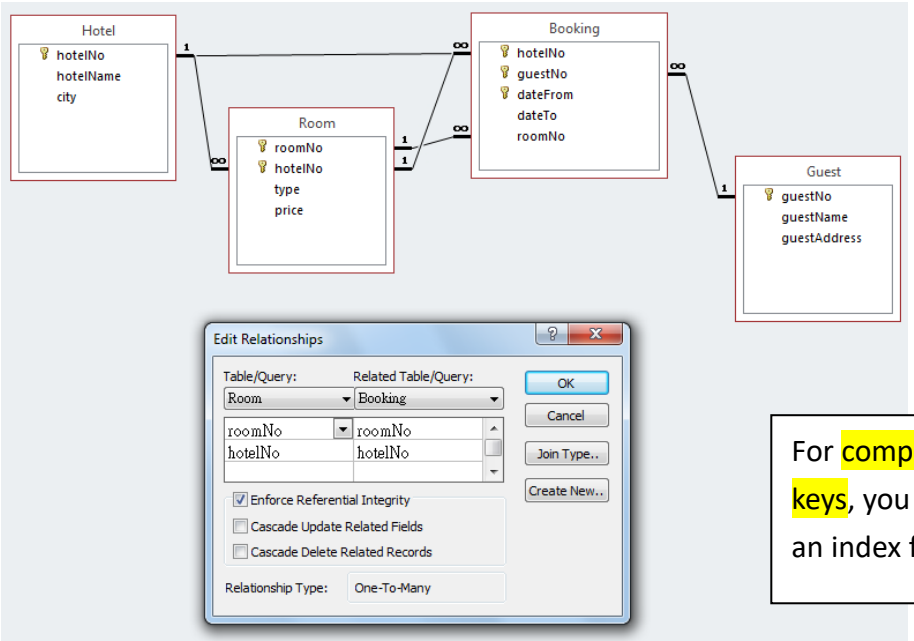
To get details of type & price

Booking contains details of bookings and (hotelNo, guestNo, dateFrom) forms the **composite** primary key. **Note that this choice of primary key tells us that:**

- For each hotel, on a given date, a guest can just book one room from it.
- But on each given date, a guest can book more than one rooms if they are in different hotels (because otherwise, the smallest set of attributes to uniquely identify a record will simply be guestNo and dateFrom)
- A practical solution will be creating a surrogate key of booking number to be used as primary key.

Although **hotelNo, dateFrom and roomNo** is also the candidate key, the problem is that during booking, usually the roomNo is unknown yet, meaning it might be null. Hence, it cannot be chosen as the primary key because of entity integrity rule

1. Identify the foreign keys in this schema.



For composite Foreign keys, you need to create an index first.

So, there are altogether 4 foreign keys.

2. Explain how the entity and referential integrity rules apply to these relations.

Entity integrity: For each of the given relations, the primary key values cannot contain any nulls in order to satisfy entity integrity rule.

- For Hotel relation, hotelNo cannot be null;
- For Room relation, both roomNo and hotelNo must contain values;
- For Guest relation, guestNo cannot be null;
- For Booking relation, none of hotelNo, guestNo, dateFrom can be null.

Referential integrity:

Room is related to Hotel through the attribute hotelNo. Therefore, hotelNo in Room should either be null or contain the number of an existing hotel in the Hotel relation. In this case study, it would probably be unacceptable to have a hotelNo in Room with a null value because hotelNo in Room is part of the composite primary key.

Booking is related to Hotel through the attribute hotelNo. Therefore, hotelNo in Booking should either be null or contain the number of an existing hotel in the Hotel relation. In this case study, it would probably be unacceptable to have a hotelNo in Booking with a null value because hotelNo in Booking is part of the composite primary key.

Booking is related to Guest through the attribute guestNo. Therefore, guestNo in Booking should either be null or contain the number of an existing guest in the Guest relation. In this case study, it would probably be unacceptable to have a guestNo in Booking with a null value because guestNo in Booking is part of the composite primary key.

Booking is also related to Room through the attributes (roomNo, hotelNo). Therefore, (roomNo, hotelNo) in Booking should either be null or contain the number of an existing (roomNo, hotelNo) in the Room relation. In this case study, it would probably be unacceptable to have hotelNo in Booking with a null value because it forms part of the composite primary key.